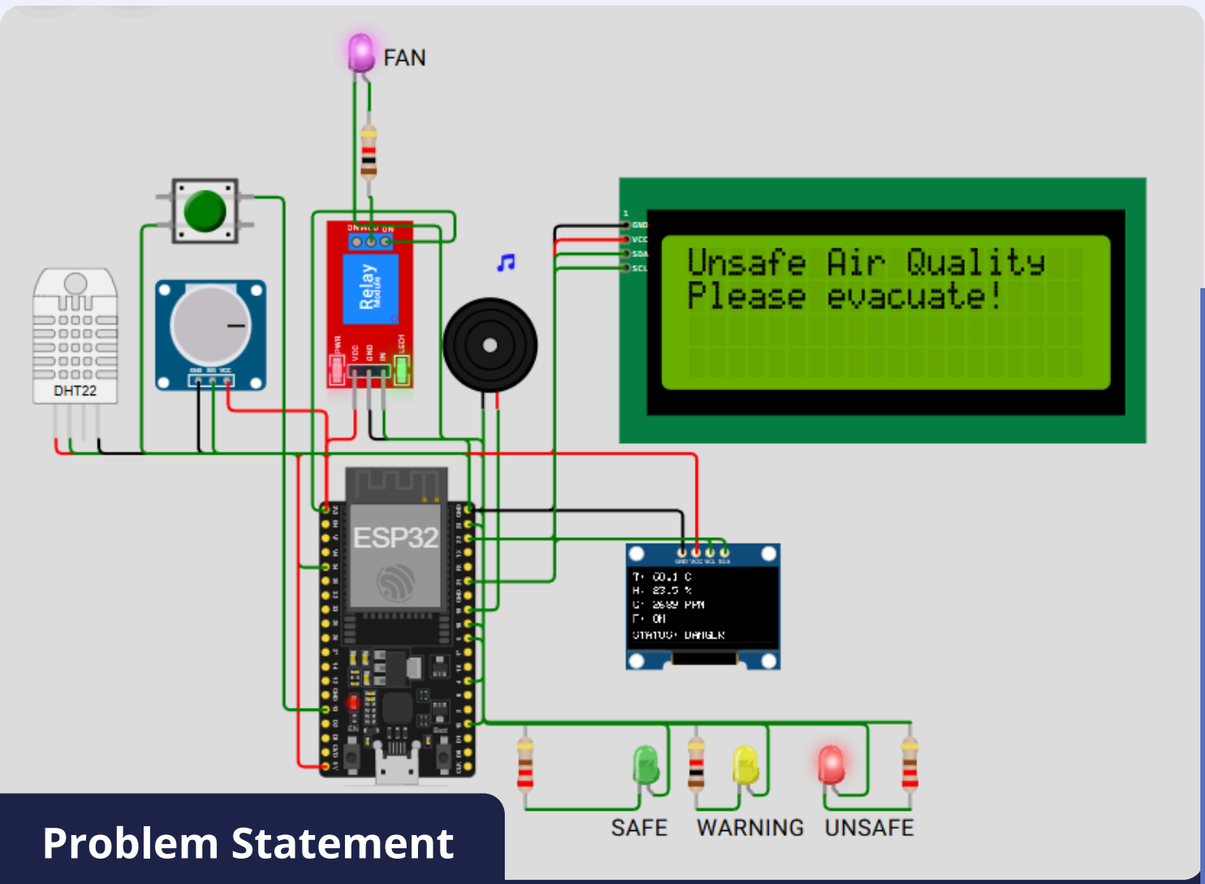
Smart Air Quality Monitoring System

CDE2243 – Internet of Things

Habiba Hassan Nur Hassan - AIU23032007 | Abdulrahman Alamodi - AIU22102263 | Mohanad Dalol - AIU22102120

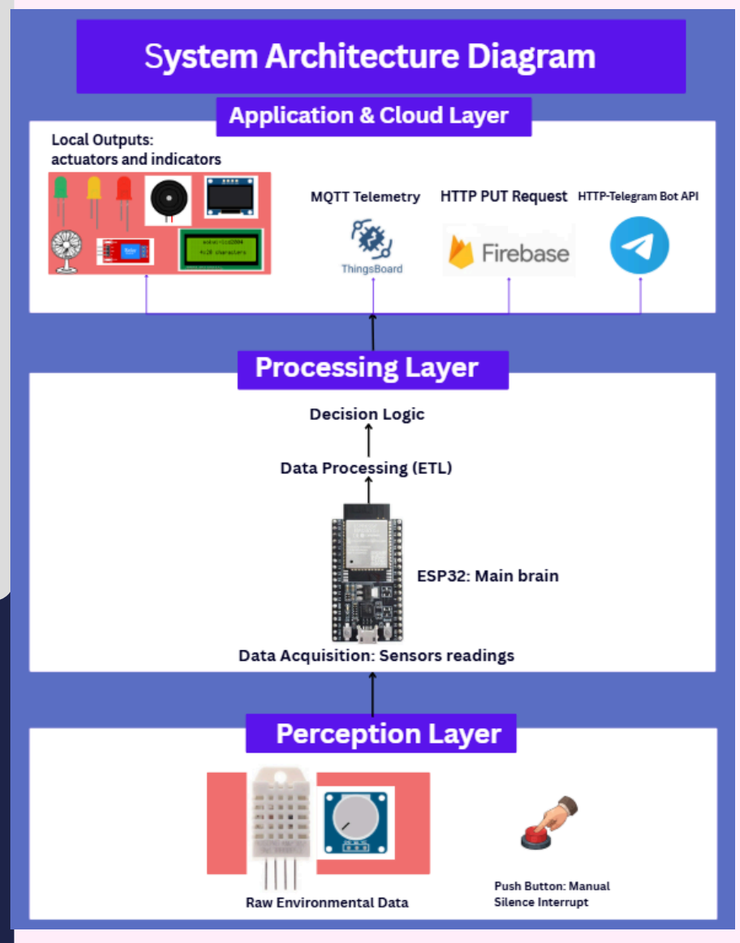


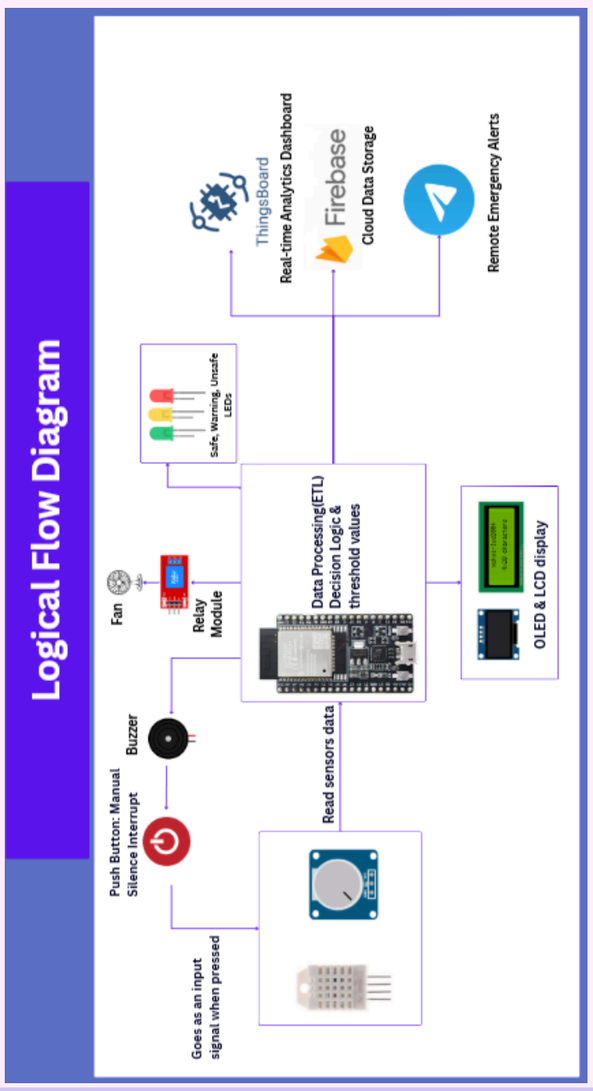
Problem Statement

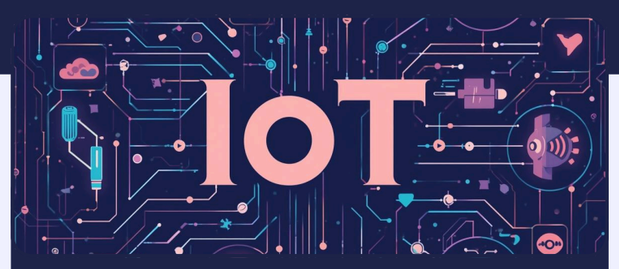
Indoor air pollution, gas leaks, and poor environmental conditions pose serious health and safety risks.
Most traditional monitoring systems lack real-time alerts, remote access, and automated responses.
A low-cost, IoT-based solution is required to continuously monitor air quality and notify users immediately when unsafe conditions occur.

System Architecture

The system consists of an ESP32 microcontroller connected to environmental sensors and actuators. Sensor data is processed locally and transmitted to cloud platforms using HTTP and MQTT protocols. The system provides real-time monitoring, automated responses, and remote alerts.







Objectives

- Monitor temperature, humidity, and gas levels in real time
- Classify air quality as SAFE, WARNING, or UNSAFE
- Automatically control actuators based on conditions
- Send alerts and visualise data on cloud platforms

Hardware & Sensor Configuration

- ESP32 DevKit microcontroller
- DHT22 temperature and humidity sensor
- Gas sensor (analog input, simulated in Wokwi)
- Relay-controlled ventilation fan
- Buzzer and LED indicators
- LCD (20×4) and OLED displays
- Push button for alarm acknowledgement

Cloud & Data Pipeline Architecture

Data Flow & Sample Logs

Sensor readings are collected every cycle and processed locally. Data is formatted as JSON and transmitted to cloud platforms. System behaviour and state transitions are logged for monitoring and debugging.

```
--- IoT System Booting ---
T: -8.8 | H: 23.5 | G: 0 | Fan: OFF | Status: SAFE
T: -8.8 | H: 23.5 | G: 1963 | Fan: ON | Status: WARN
T: -8.8 | H: 23.5 | G: 2447 | Fan: ON | Status: WARN
T: 68.1 | H: 23.5 | G: 2689 | Fan: ON | Status: DANGER
```

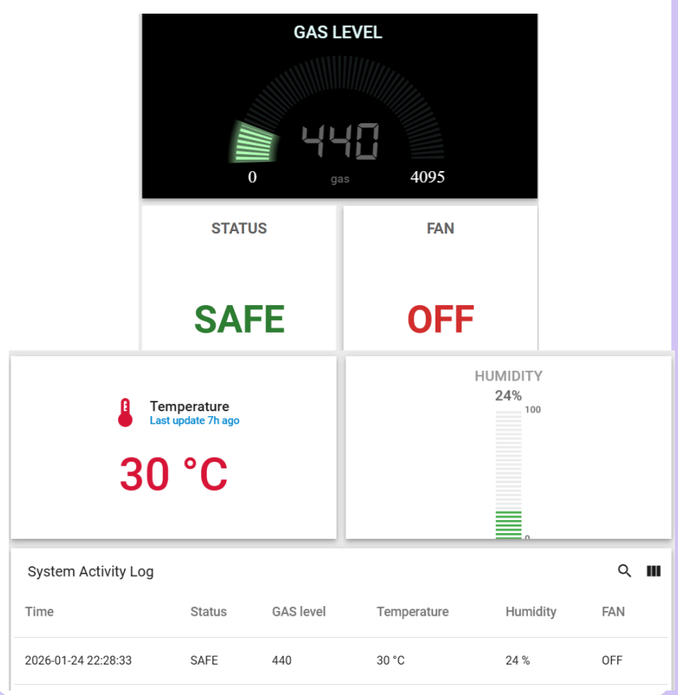
Analytics & Decision Logic

The system applies threshold-based decision logic to classify air quality. Gas sensor readings are first aggregated using a moving average to reduce noise. Based on predefined thresholds, the system determines the air quality state and triggers appropriate actions.

Decision Logic:

- SAFE: All parameters within safe limits
- WARNING: Moderate gas, temperature, or humidity levels
- UNSAFE: High gas concentration or critical temperature/humidity

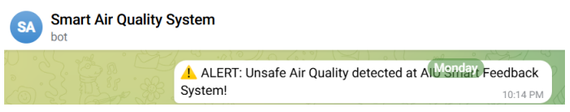
Dashboard & Visualisation



Results & Observations

The system successfully detects unsafe environmental conditions and responds autonomously.
When gas concentration or environmental parameters exceed thresholds:

- The ventilation fan is activated
- Visual and audible alerts are triggered
- Notifications are sent via Telegram
- Data is updated in real time on cloud platforms



Security & Privacy Considerations

- The system uses secure communication protocols and token-based authentication.
- Sensitive credentials are not exposed publicly.
- Only environmental data is collected, and no personal user data is stored.

Conclusion & Future Enhancements

The proposed IoT air quality monitoring system effectively integrates sensing, cloud communication, analytics, and alerting. It provides a reliable and scalable solution for real-time environmental monitoring.

Future enhancements include:

- Additional environmental sensors
- Mobile application integration

Demo Links

- https://wokwi.com/projects/454227171978886145
- https://smart-air-quality-system-97e90-default-rtdb.firebaseio.com
- https://thingsboard.cloud/dashboard/ac03abb0-f935-11f0-9817-8bc7e0ad4d62?publicId=6cf08b80-ff22-11f0-8287-999f1041c10d

References

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